Geophys. J. Int. (1999) **139,** 447–463

Clockwise rotations recorded in Early Cretaceous rocks of South Korea: implications for tectonic affinity between the Korean Peninsula and North China

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Accepted 1999 June 15. Received 1999 June 15; in original form 1998 December 5

SUMMARY

Recent interest has focused on whether South Korea may have undergone variable tectonic rotations since the Cretaceous. In an effort to contribute to the answer to this question, we have completed a palaeomagnetic reconnaissance study of Early Cretaceous sedimentary and igneous rocks from the Kyongsang basin in southeast Korea. Stepwise thermal demagnetization isolated well-defined characteristic magnetization in all samples. The palaeomagnetic directions reveal patterns of increasing amounts of clockwise (CW) rotation with increasing age for Aptian rock units. Palaeomagnetic declinations indicate clockwise vertical-axis rotations of $R = 34.3^\circ \pm 6.9^\circ$ for the early Aptian rock unit, $R = 24.9^\circ \pm 10.6^\circ$ for the middle Aptian, and $R = -0.9^\circ \pm 11.8^\circ$ for the late Aptian relative to eastern Asia. The new Cretaceous palaeomagnetic data from this study are consistent with the hypothesis that Korea and other major parts of eastern Asia occupied the same relative positions in terms of palaeolatitudes in the Cretaceous. An analysis of and comparison with previously reported palaeomagnetic data corroborates this hypothesis and suggests that much of Korea may have been connected to the North China Block since the early Palaeozoic. A plausible cause of the rotation is the westward subduction of the Kula plate underneath the Asian continent, which is inferred to have occurred during the Cretaceous according to several geological and tectonic analyses.

Key words: Korea, North China, palaeomagnetism, rotation.

1 INTRODUCTION

The collision between the North and South China blocks (NCB and SCB) has played a central role in shaping the eastern Asian continent. One early tectonic model of the collision, based on palaeomagnetic data and geological evidence (Zhao & Coe 1987), proposed that the North and South China blocks started colliding in the latest Permian and Early Triassic near the eastern end of what is now the Qinling mountain belt (Fig. 1); this was followed by counterclockwise rotation of the NCB with respect to the SCB until final suturing was completed during mid-late Jurassic (Zheng *et al.* 1991; Huang & Opdyke 1991; Gilder *et al.* 1993; Zhao *et al.* 1996; Gilder & Courtillot 1997).

Since the main collision and suturing between the NCB and SCB, however, there has been significant tectonic activity in the region. To the north, the scissor-like closure of the United China Block (NCB, SCB and Mongolia) and Siberia did not end until Juro-Cretaceous times (Zhao *et al.* 1990, 1996; Gilder & Courtillot 1997). To the east, the subduction of the Kula plate beneath the Asian continental margin and the opening of the Bohai Sea began in the Early Cretaceous and continued to Cenozoic times (Maruyama *et al.* 1989; Kong *et al.* 1997). Widespread Late Mesozoic to recent remagnetizations, which are probably related to these events, have been identified in eastern China (Kent *et al.* 1987; Zhao 1987; Huang & Opdyke 1996), Korea (Otofuji *et al.* 1989; Doh & Piper 1994; Doh *et al.* 1997), and Siberia (Halim *et al.* 1998).

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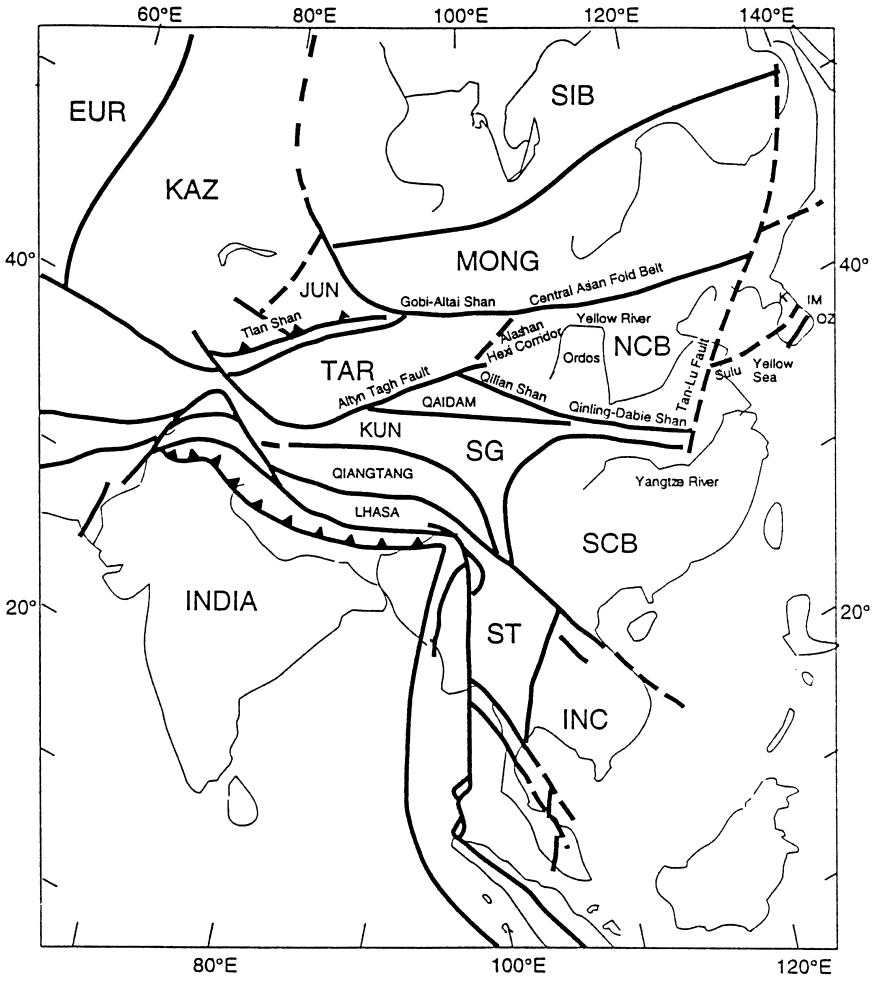


Figure 1. Sketch map of the present locations of the continental blocks of China and Southeast Asia showing the main sutures and faults. The major blocks are EUR: Europe; KAZ: Kazakhstan; KUN: Kunlun; SIB: Siberia; MONG: Mongolia; JUN: Junggar; TAR: Tarim; NCB: North China Block; SCB: South China Block; Qiangtang (North Tibet); Lhasa (South Tibet); K: Korea; SG: Songpan-Ganzi; ST: Shan Thai; and INC: Indochina. Abbreviations: K, Korea; IM, Imjingang fault; OZ, Okchon zone. Modified after Enkin *et al.* (1992).

The Korean Peninsula (Fig. 1) is situated east of the North China Block, and its affinities with both the NCB and SCB to the west are intensely debated. Traditionally, Korea has been incorporated with the NCB into a single craton (Sino-Korea) based on the close similarity of Palaeozoic stratigraphies (Reedman & Um 1975; Lee 1987, pp. 231–234; Chang 1995). Many recent assessments of the geology of eastern Asia, however, have highlighted the importance of the Tanlu fault in eastern China and the Imjingang and Okchon (also spelled as Ogcheon) zones of Korea as possible eastern boundaries of the North China Block (Fig. 1). These NE–SW-trending faults have been considered as part of a large left-lateral strike-slip fault system extending throughout eastern China, Korea, and up to the Sea of Okhotsk (Xu *et al.* 1987; Xu *et al.* 1993; Yanai *et al.* 1993). Depending on the significance of these fault movements, it is possible that the Korean Peninsula (or part of it) was formerly located close to the South China Block, a link which has been advocated by several workers (Lee *et al.* 1987; Otofuji *et al.* 1989; Gilder *et al.* 1995; Yin & Nie 1996; Lee *et al.* 1996). In order to reconcile differences in both geological and palaeomagnetic data from Korea, several workers also divided Korea into different blocks, such as the North and South Korea blocks (Lee *et al.* 1987) and the North and South Sino-Korean blocks (Lee *et al.* 1997), each having its own tectonic history. The lack of agreement about the tectonic coherence of Korea hinders our understanding of the regional tectonic evolution and palaeogeography.

Palaeomagnetism remains a principal tool not only to constrain the geological and tectonic history of crustal blocks and test the validity of various models, but also to aid in the recognition of vertical-axis block rotations. Vertical-axis rotations have been shown to be a common consequence of crustal deformation in settings with strike-slip and/or thrusting motions (e.g. Beck 1976; Coe *et al.* 1985; Luyendyk *et al.* 1985). Over the last decade, there has been substantial progress in palaeomagnetic research in Korea (Ito & Tokieda 1980; Min *et al.* 1993; Kim & Jeong 1986; Kim *et al.* 1992; Otofuji *et al.* 1983, 1986, 1989; Lee *et al.* 1987; Shibuya *et al.* 1988; Kim & Van der Voo 1990; Doh & Piper 1994; Lee *et al.* 1996, 1997; Doh *et al.* 1997; Kang & Kim 1998). The results constitute a beginning and provide a reference for comparison and refinement.

A few Cretaceous results are also available and suggest variable tectonic rotations for the Korean blocks (Kim & Jeong 1986; Lee *et al.* 1987; Doh & Piper 1994; see Fig. 2 for the distribution of sampling localities in these studies). Kim &

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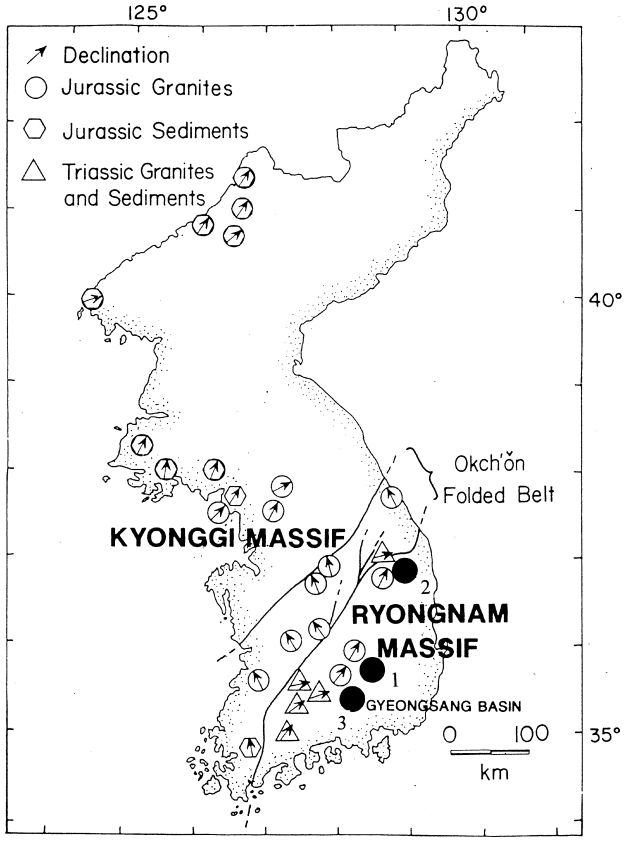


Figure 2. Simplified map of Korea showing Late Jurassic palaeomagnetic declinations and the distribution of Cretaceous sampling areas of previously published studies. 1: Taegu area (Kim & Jeong 1986; Otofuji *et al.* 1986; Lee *et al.* 1987); 2: Samcheok area (Doh & Piper 1994); 3: seven localities in the south and southwest of the Kyongsang basin (Kang & Kim 1998). Figure modified after Kim & Van der Voo (1990).

Jeong (1986) studied Cretaceous rocks from the Taegu area and concluded that this part of South Korea has undergone a total of 38.2° clockwise rotation relative to Eurasia ($27.4° \pm 14.9°$ during the Early Cretaceous, and $10.8° \pm 10.6°$ in the Late Cretaceous, where the plus or minus value is the 95 per cent confidence interval of the mean). Lee *et al.* (1987) suggested a clockwise rotation of $9° \pm 11°$ of southern Korea relative to the Chinese blocks in post-Cretaceous times, whereas Doh & Piper (1994) argued that the Korea Block has undergone a local clockwise rotation of $\sim 30°$ which occurred between Cretaceous and recent times. A recent report by Kang & Kim (1998) also indicates that South Korea has experienced a clockwise rotation ($6.5° \pm 9.2°$) relative to the Chinese blocks since the Early Cretaceous. Thus, although the palaeomagnetic data are limited, they generally support the contention that southern Korea may have rotated clockwise relative to the Chinese blocks in Cretaceous times. However, several workers have argued that there is no geological evidence indicative of these rotations (Lee 1987; Lee *et al.* 1996; p. 282). Thus, the basic question of whether or not tectonic rotation has occurred remains in contention.

In this paper, we present new palaeomagnetic results obtained from Cretaceous rocks in southeast Korea in an effort to constrain the kinematics and timing of possible tectonic rotations. Based on our new results and an analysis of and comparison with previously reported palaeomagnetic data, we suggest that a $\sim 30°$ clockwise rotation of southeast Korea may be a characteristic feature in the Late Mesozoic tectonics of the Korean Peninsula. We then attempt a palaeogeographic reconstruction of the East Asian margin in the Early Cretaceous. In particular we propose that much of Korea may have been connected to the North China Block.

2 REGIONAL GEOLOGICAL AND TECTONIC FRAMEWORK

Geologically, South Korea is divided (from north to south) into four provinces: the Kyonggi Massif, the Okchon zone (foldbelt), the Ryongnam Massif, and the southeasterly Kyongsang (Gyeongsang) basin (Fig. 2). The geology of Korea has been summarized in a comprehensive book (Lee 1987). A number of detailed reviews of Korean geology and tectonic maps are also available (e.g. Chang 1995). For the purpose of this paper, we outline only those aspects that are relevant to our discussion of the palaeomagnetic data.

As shown in Fig. 2, the Okchon zone extends in a NE–SW direction diagonally across the peninsula with a width of about 80 km. The belt was a site of Late Jurassic and Early Cretaceous deformation (the Daebo Orogeny, see Chang 1995) and contains possible ophiolites (Lee 1987). This belt is divided into two parts, the southwesterly metamorphosed zone and the northeasterly non-metamorphic sedimentary zone. The former is composed of metamorphic Precambrian and Palaeozoic strata intruded by Jurassic Daebo granites and is also called the Honam shear zone. Recent advances in geological interpretation of the 'Okchon nappe' in this zone suggest that the nappe underwent major tectonic transport from the South China Block (Chang 1995). In contrast, the northeasterly non-metamorphosed zone of the Okchon fold belt has less well-defined boundaries. Fossils found in the Palaeozoic sediments have been correlated with those from North China (Lee 1987; Chang 1995). Previously published palaeomagnetic results by Kim & Van der Voo (1990) suggest that a large part of the Okchon belt may have undergone counterclockwise rotations with respect to areas both north and south of the belt in Late Jurassic to Early Cretaceous times (Fig. 2).

The Kyongsang basin occupies the southeastern-most part of South Korea. The basal unit of the Kyongsang basin, the Myogok Formation, is exposed in a small area in the northwest corner of the basin (Chang 1988) and is assigned to a late Upper Jurassic age based on fresh-water mollusks (Yang 1984). The Cretaceous Kyongsang supergroup overlies the Myogok Formation with an angular unconformity due to Nakdong folding, which was related to the third phase of the Yanshan Orogeny (Late Jurassic) in eastern China and the Oga phase of the Sakawa tectonic cycle in Japan (Chang 1988). The formation of the basin was facilitated by extensional tectonics, and the basin was intruded by the Pulguksa granite group (100–73 Ma) in the north.

The Cretaceous strata of the Kyongsang basin are dominated by unconformity-bounded units of non-marine sedimentary and igneous rocks (the Kyongsang supergroup). The Kyongsang supergroup is divided into three groups: the Sindong, Hayang, and Yuchon groups in ascending order (Fig. 3). The Sindong group consists of sediments of a prevolcanic phase, the Hayang group comprises non-volcanic sediments with some volcanic horizons, and the Yuchon group consists of volcanic formations.

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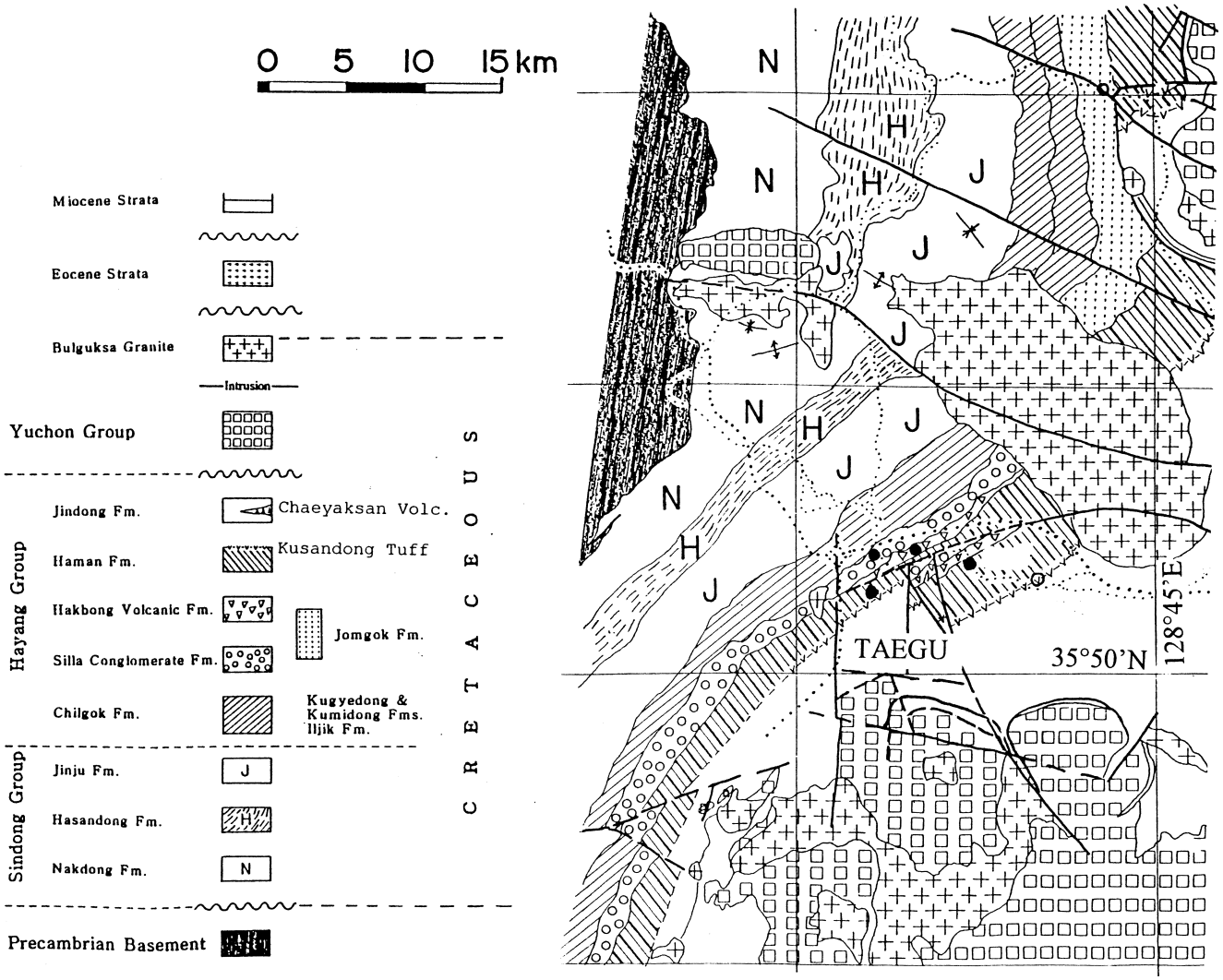


Figure 3. Simplified geological map of Taegu area, showing the structural trends of the Cretaceous formations. Black dots represent sampling areas for this study. Modified after Chang & Park (1995). N: Nakdong Formation; H: Hasandong Formation; J: Jinju Formation.

Of the seven formations within the Sindong and Hayang groups, four of them (the Haman, Silla, Chilgok and Hasandong formations) contain red beds of mudstone and siltstones, whereas two of the others (Jindong and Jinju formations) consist mainly of black shale and one (Nagdong Formation) is dark grey mudstone. These formations unconformably overlie the Precambrian gneisses and Jurassic granites and crop out in a NNE-trending belt (Fig. 3).

3 PALAEOMAGNETIC SAMPLING AND LABORATORY TECHNIQUES

We collected a total of 107 samples of Early Cretaceous igneous and sedimentary rocks at 15 sites from the Hayang group in the Kyongsang basin (Fig. 3 and Table 1), with age span roughly between 125 and 114 Ma (i.e. within the Aptian). The age estimates performed by Korean geologists (e.g. Chang & Park 1995) and other workers are mainly based on Charophyte determinations coupled with radiometric dating on several key samples (Otofuji *et al.* 1983; Lee *et al.* 1987; Chang *et al.* 1998). The most important and recent dating comes from the Kusandong tuff, which is a tuff horizon on the uppermost part of the Haman Formation in the city of Taegu. Zircon from this tuff layer gives a U–Pb age of 113.6 ± 10 Ma, which roughly coincides with the Aptian/Albian boundary. Sampling sites were located in fault-bounded regions with slightly variable attitudes. The Early Cretaceous strata are gently tilted but not folded, so corrections for a plunging fold axis are not needed. Since tilting affects rocks as young as Late Cretaceous in the Kyongsang basin, tilting is probably latest Cretaceous or Early Tertiary. Distributed samples were drilled at each site with a gasoline-powered drill and orientated *in situ* with both magnetic and sun compasses. The mean difference between the two compass readings is $\pm 1^\circ$, in excellent agreement with the local geomagnetic field declination predicted from the 1995 International Geomagnetic Reference Field (IGRF) for the region (-7.2°), indicating that local magnetic anomalies are moderate and averaged out in the mean. Stratigraphic locations and orientations of each sampling site were carefully recorded during the course of the fieldwork. Field samples were trimmed into 2.2 cm long cylinders for subsequent palaeomagnetic analysis.

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Table 1. Palaeomagnetic sampling details of Early Cretaceous sites in South Korea from this study.

Site ID	Age	Rock Type (Formation)	Slat (N°)	Slong (E°)	N	Strike	Dip
Silla.01	A1	Red sandstone (Silla)	35.9	128.6	5	63	18SE
Silla.02	A1	Red sandstone (Silla)	35.9	128.6	6	63	18SE
Silla.03	A1	Red sandstone & volcanic pebble (Silla)	35.9	128.6	5	63	18SE
Silla.04	A1	Red sandstone & volcanic pebble (Silla)	35.9	128.6	5	63	18SE
Hakbo.01	A2	Volcanics (Hakbong)	35.9	128.6	4	66	16SE
Hakbo.02	A2	Volcanics (Hakbong)	35.9	128.6	2	73	16SE
Hakbo.03	A2	Volcanics (Hakbong)	35.9	128.6	6	66	16SE
Haman.01	A3	Redbeds (Haman, lower)	35.9	128.6	5	140	9.5SW
Haman.02	A3	Redbeds (Haman, lower)	35.9	128.6	5	140	9.5SW
Haman.03	A3	Redbeds (Haman, lower)	35.9	128.6	4	140	9.5SW
Haman.04	A3	Redbeds (Haman, lower)	35.9	128.6	6	140	9.5SW
Haman.05	A3	Redbeds (Haman, upper)	35.9	128.6	6	63	11SE
Haman.06	A3	Redbeds (Haman, upper)	35.9	128.6	6	63	11SE
Haman.07	A3	Redbeds (Haman, upper)	35.9	128.6	4	63	11SE
Haman.08	A3	Redbeds (Haman, upper)	35.9	128.6	5	64	10.5SE
Haman.09	A3	Redbeds (Haman, upper)	35.9	128.6	5	64	10.5SE

Explanation: Age—approximate age of rocks studied, based mainly on palaeontological and geological evidence. A = Aptian, 1 = Early, 2 = Middle, 3 = Late. SLat/SLong: latitude/longitude of sampling locality. N: number of samples used.

All the experimental work was undertaken in a magnetically shielded room. The samples were subjected to progressive thermal demagnetization and measured at each stage of treatment with a 2G cryogenic magnetometer at the palaeomagnetic laboratory of the University of California, Santa Cruz. Bulk magnetic susceptibility was also measured after every demagnetization step to detect whether chemical changes were affecting the magnetization during progressive heating. Magnetization directions were determined by principal component analysis (Kirschvink 1980), the distribution of palaeomagnetic directions at each site was calculated using Fisher (1953) statistics, and site mean directions of all demagnetized data were derived by giving unit weight to each mean sample direction. A few representative samples were also selected for a set of rock magnetic measurements to examine their mineralogical characteristics. These rock magnetic measurements were performed at the Institute for Rock Magnetism, University of Minnesota, and include (1) high-field (1 T) Curie temperature determinations, (2) low-temperature (10 K) cycling of saturation isothermal remanent magnetization, and (3) measurement of hysteresis loop parameters.

4 PALAEOMAGNETIC AND ROCK MAGNETIC RESULTS FROM THIS STUDY

4.1 Palaeomagnetic results

Early, middle, and late Aptian formations were investigated (Table 2) in this study. Their results are described below sequentially from old to young.

4.1.1 Early Aptian Silla Formation

We sampled four reddish sandstone, siltstone, and conglomerate localities in the vicinity of Kumhodong, northwest Taegu. The red sandstones may include the upper member of the Chilgok Formation because the boundary between the Chilgok and Silla formations is arbitrary (Chang 1988). The NRM intensities of the red sandstone in this formation range from 1 to 11 mA m^{-1} (typically $\sim$6 mA m^{-1}). The directions of a soft component during the initial demagnetization (to 200 °C) generally cluster near the geocentric axial field direction (GAD) at the sampling area. Progressive thermal demagnetization to 700 °C, however, reveals two other components of magnetization.

An intermediate unblocking temperature component (ITC) was isolated in all samples by best-fitting lines to demagnetization data between 300 and 500 °C. Interestingly, the directions of the ITC in these red sandstones are reversed with a mean *in situ* direction at D (Declination) = 169.2°, I (inclination) = −55.8°, $N = 6$, k (precision parameter of site mean direction) = 16.0, and $\alpha 95$ (95 per cent confidence circle of site mean direction) = 17.3°. This reversed magnetization is compatible with a reversed event prior to the Brunhes (>0.78 Ma) and is incompatible with a Holocene field direction.

A high unblocking temperature component (HTC) was isolated between 625 and 700 °C in these red sandstone samples (Fig. 4 and Table 2). The HTC has normal polarity and is interpreted as the characteristic remanent magnetization (ChRM) for the Silla Formation on the basis of linear trajectories of demagnetization towards the origin and a similar direction from sample to sample. Before tilt correction, the ChRM displays a north-northeasterly direction with intermediate downward inclination ($D = 23.9°$, $I = 53.9°$, $N = 4$ sites, $k = 286.6$, $a95 = 5.4°$); after tilt correction, it is orientated northeast with a steeper inclination ($D = 50.2°$, $I = 62.0°$, $k = 286.6$, $a95 = 5.4°$). The normal polarity at all four sites may correspond to the Cretaceous Normal Superchron, which is in agreement with the age of the formation. Fig. 5 shows the directions of magnetization in conglomerate pebbles derived from volcanics in the Silla sites 03 and 04. The red siltstone samples yield the same directions as those of pure Silla redbed, whereas the directions of the volcanic pebbles are randomly orientated. This would suggest that the magnetization of the redbeds has been stable since the formation of the conglomerate and that there was no event that remagnetized the entire Silla conglomerate Formation.

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Table 2. Summary of characteristic palaeomagnetic results from South Korea.

Site ID	Age	R/N	Range (°C)	G_Dec	G_Inc	S_Dec	S_Inc	S_Long	S_Lat	a95	k2
Early Aptian											
This study, Kumhodong area (128.6°E, 35.9°N), Silla Redbeds and Conglomerate Formation											
Silla.01	A1	0/5	625–700	16.8	52.9	41.1	63.2	188.8	57.4	10.5	54.1
Silla.02	A1	0/6	500–700	20.4	55.7	48.4	64.7	186.4	52.6	4.7	201.5
Silla.03	A1	0/3	625–700	27.8	50.0	50.5	57.4	193.7	52.6		11.2
#Basaltic pebble 35A			200–350	343.8	37.9	348.1	55.5	42.9	80.4		
#Granite pebble 36A			600–700	263.2	−5.3	263.8	1.1	42.7	−4.7		
Silla.04	A1										
#Red siltstone 37A			600–675	30.8	56.4	60.2	62.0	190.1	44.0		
#Pink granite 38A			600–675	29.2	−15.5	26.5	−5.1	270.1	44.3		
#Rhyolite sample 39A			500–700	101.2	−13.6	96.2	−24.2	225.4	−12.4		
#Rhyolite sample 40B			600–700	103.9	−16.8	98.0	−27.9	226.2	−15.0		
Mean HTC	N = 15 samples			21.3	53.8	47.3	62.7	188.9	53.7	5.7	45.7
Mean HTC	N = 4 sites			23.9	53.9	50.2	62.0	191.7	51.2	5.4	286.6
Reversed ITC component from Silla Formation											
	N = 6 samples		100–500	169.2	−55.8	184.3	−72.4	315.3	−67.1	17.3	16.0
Otofuji *et al.* (1986), Taegu (128.5°E, 36.0°N), Jinju, Chilgok and Nagdong Formations											
Mean*	Al	N = 3	(k2/k1 = 6.51)	10.3	59.3	36.9	63.7	187.9	60.7		436.5
Lee *et al.* (1987), Taegu (128.5°E, 36.0°N), Jinju, Chilgok and Silla Formations (sites 3–9 in the original publication)											
Mean*	A1	N = 7	(k2/k1 = 1.41)	13.9	53.0	37.9	61.3	193.2	60.2	5.8	107.7
Doh and Piper (1994), Samcheok Coalfield (129.0°E, 37.1°N), Overprinted direction											
Mean	Al	N = 14		56.1	54.8	56.1	54.8	200.4	46.7	9.6	18.1
Average with Otofuji *et al.* (1986), Lee *et al.* (1987), and Doh & Piper (1994)											
N = 4 studies			(k2/k1 = 3.84)	26.1	56.6	45.9	60.7	193.8	54.8	6.9	176.0
Middle Aptian											
This study, Taegu area (128.6°E, 35.9°N), Hakbong Volcanic Formation											
Hakbo.01	A2	0/6	500–700	18.0	49.4	35.5	59.6	198.2	62.1	4.5	220.6
Hakbo.02	A2	0/3	500–700	44.6	27.7	53.6	32.3	225.7	41.2	44.6	8.7
Hakbo.03	A2	0/3	500–700	18.2	53.6	39.2	63.5	189.0	59.2	6.3	382.4
Mean N = 3 flows				29.1	44.4	44.9	52.2	207.3	54.6	28.0	20.4
By samples N = 12				26.0	46.0	42.5	54.6	204.8	57.4	10.1	19.6
Lee *et al.* (1987), Taegu area (128.6°E, 35.9°N), Hakbong Volcanic Formation											
11	A2	17		16.1	51.6	26.1	58.8	200.7	69.1	4.4	65.6
12	A2	17		2.3	53.7	22.6	58.0	203.1	71.9	1.7	432.4
13	A2	14		10.4	55.6	46.8	55.7	203.8	52.6	5.6	52.0
Mean*	A2	N = 3	(k2/k1 = 1.00)	9.7	53.8	32.0	58.0	202.8	64.6	11.1	124.8
Average with Lee *et al.* (1987)											
N = 6 flows		(k2/k1 = 1.20)		20.2	49.5	38.9	55.3	205.4	59.6	11.1	37.5
Late Aptian											
This study, Taegu area (128.6°E, 35.9°N), Haman Red Sandstone and Siltstone Formation											
Lower Section (Sankyok-dong)											
Haman.0.1	A3	0/5	625–675	6.7	58.0	363.5	64.2	102.6	78.2	9.8	61.5
Haman.0.2	A3	0/5	350–675	6.7	52.6	356.2	58.9	95.2	85.0	6.0	162.8
Haman.0.3	A3	0/4	350–650	5.2	53.6	354.1	59.7	86.8	83.3	5.3	298.4
Haman.0.4	A3	0/6	350–650	13.6	59.4	1.0	66.4	131.8	76.6	5.3	161.9
Mean N = 20 samples				8.3	56.2	356.4	62.7	111.6	80.9	3.1	114.0
Mean N = 4 sites				7.9	55.9	356.0	62.3	108.8	81.5	4.4	433.8
Mean of ITC N = 20 samples				358.2	54.5	345.8	59.5	69.1	78.0	5.0	44.1
Upper section (Kyungpook National University campus)											
Haman.05	A3	0/6	625–700	15.0	40.8	22.8	48.5	228.8	69.7	8.2	68.2
Haman.06	A3	0/6	600–700	9.3	42.8	17.0	51.3	227.2	75.3	12.1	31.7
Haman.07	A3	0/4	500–675	9.7	39.8	16.5	48.2	238.8	74.6	13.0	50.9
Haman.08	A3	2/3	560–700	37.6	44.8	48.0	48.6	212.8	50.8	13.6	32.4
Haman.09	A3	1/4	625–675	24.5	51.0	36.6	56.8	203.3	62.3	13.7	32.3
	Normal	20/23		24.7	51.0	36.8	56.8	203.4	62.0	10.2	35.9
	Reversed	3/23		224.3	−39.9	233.0	−42.7	36.8	−43.7	14.0	79.1
Mean N = 23 samples				18.6	44.3	27.7	51.3	218.8	67.2	5.2	31.1
Mean N = 5 sites				18.8	44.3	28.2	51.4	219.5	66.3	9.2	70.4
Average lower and upper section results											
		N = 9 sites		14.6	49.6	16.2	57.2	202.2	77.7	8.3	39.3

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Table 2. (*Continued.*)

Site ID	Age	R/N	Range (°C)	G_Dec	G_Inc	S_Dec	S_Inc	S_Long	S_Lat	a95	k2
Lee *et al.* (1987), Taegu (128.6°E, 35.9°N), Haman Formation (sites 14+ − 15 in the original publication)											
Mean*	A3	N = 3		23.7	43.9	34.5	47.6	221.3	60.1		31.4
(k2/k1 = 1.03) with a positive reversal test result											
Otofuji *et al.* (1986), Taegu area (128.5°E, 36.0°N), Haman Formation											
Mean	A3	N = 6	(k2/k1 = 5.99)	9.7	64.0	27.6	59.5	198.0	68.0	3.9	285.3
Average with Otofuji *et al.* (1986) and Lee *et al.* (1987)											
N = 3 localities			(k2/k1 = 1.82)	17.0	52.6	26.7	55.0	210.2	68.6		95.4
Overall mean for the Aptain aged strata in South Korea											
N = 9 studies			(k2/k1 = 1.70)	21.6	53.7	37.4	57.9	199.5	61.2	6.6	96.7

Explanation. Age: approximate age of rocks studied, based mainly on palaeontological and geological evidence. A: Aptian, 1: Early, 2: Middle, 3: Late. R/N: number of reversed (R) to number of normal (N) samples. G_Dec/G_Inc, S_Dec/S_Inc: declination and inclination in geographic and stratigraphic coordinates, respectively; S_Long/S_Lat: east longitude and north latitude of VGP in stratigraphic coordinates; a95/A95: radius of circle of 95 per cent confidence about the direction/VGP in degrees (stratigraphic coordinates). k2(k1)/K: Fisher (1953) precision parameter for direction after (before) tilt correction/for VGP. HTC/ITC: high/intermediate temperature component identified in this study.
volcanic pebbles used for the conglomerate test (Fig. 5).
* Recalculated data according to the age of formation.

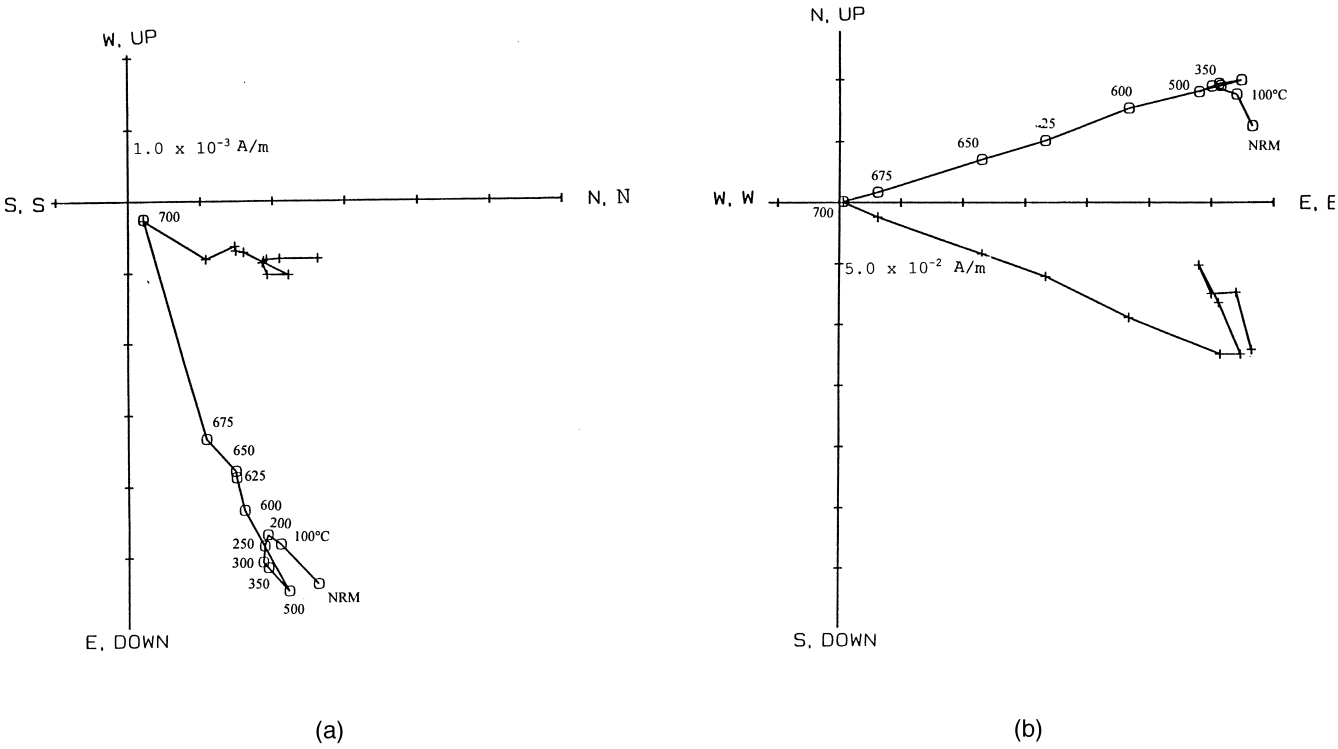


Figure 4. Representative vector plots of rocks from the Early Cretaceous Silla Formation. (a) Thermal demagnetization of red sandstone sample 95K0024B, showing an intermediate unblocking temperature component (ITC) between 300 and 500 °C. (b) Thermal demagnetization of granite pebble sample 95K0040B. Directions are plotted in geographical coordinates. Crosses: vector endpoints projected onto the horizontal plane; Circles: endpoints projected onto the north–south vertical plane; NRM: natural remanent magnetization.

4.1.2 Middle Aptian Hakbong volcanic Formation

In the Taegu area, the Hakbong volcanic Formation lies between the Silla conglomerate and Haman formations. With a maximum thickness of about 200 m and a lateral extent of only 15 km (Chang & Park 1995), it is composed of basaltic lava overlain by agglomerates and sandstones. Our sampling in this formation was limited by weather and field conditions. As a result, we only sampled three localities (covering three lava flows) of grey coloured intermediate-acidic volcanic rocks (Table 2). Radiometric dates are unavailable for these rocks, but their ages based on geological relations are inferred to be early middle Aptian, according to our field guides from the Kyungpook National University. We regard this early middle Aptian age assignment for the Hakbong Formation as reasonable because two radiometric age determinations from andesites that cut across the youngest part of the Kyongsang supergroup give a minimum age of 89 ± 4 Ma (Lee *et al.* 1987),

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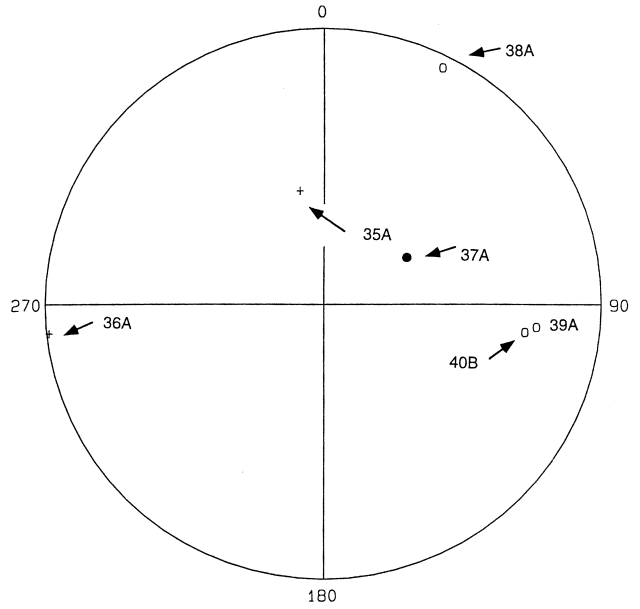


Figure 5. Equal-area projection of stable directions for samples in Site Silla.04 and two volcanic pebbles in Site Silla.03 (data with '#' in Table 2). Directions are plotted in stratigraphic coordinates. the filled circle denotes the red sandstone sample 37A. Open circles (crosses) indicate upper (lower) hemisphere projection.

and grains from the Kusandong tuff yield a U–Pb date of 113.6 ± 10 Ma, as mentioned above. We performed stepwise thermal and AF demagnetization experiments on 12 samples from the three localities. The magnetization of the Hakbong basalt samples from the Taegu area is quite stable and well behaved. A small, random component of magnetization was easily removed by demagnetization and the characteristic component quickly revealed itself (Fig. 6). Only normal polarity directions were obtained (Table 2). The overall mean direction for the three flows is $D = 29.1^\circ$, $I = 44.4^\circ$, $N = 3$ flows, $k = 20.4$ in geographical coordinates, and $D = 44.9^\circ$, $I = 52.2^\circ$, $k = 20.4$ in stratigraphic coordinates. The small number of flows leads to a much larger uncertainty in the mean direction than for the Silla Formation above. Although no fold or reversal test is available, the fact that the same characteristic direction is carried by both magnetite and haematite suggests that the haematite was probably formed by auto-oxidation processes during primary cooling of the basalts. The number of flows is not large enough to ensure averaging out of secular variation but the mean direction is not far from that for the Silla Formation and is of normal polarity, again consistent with the Cretaceous Normal Superchron and the age of the formation. Thus, we are reluctant to exclude this result, but its extremely large uncertainty must be kept in mind.

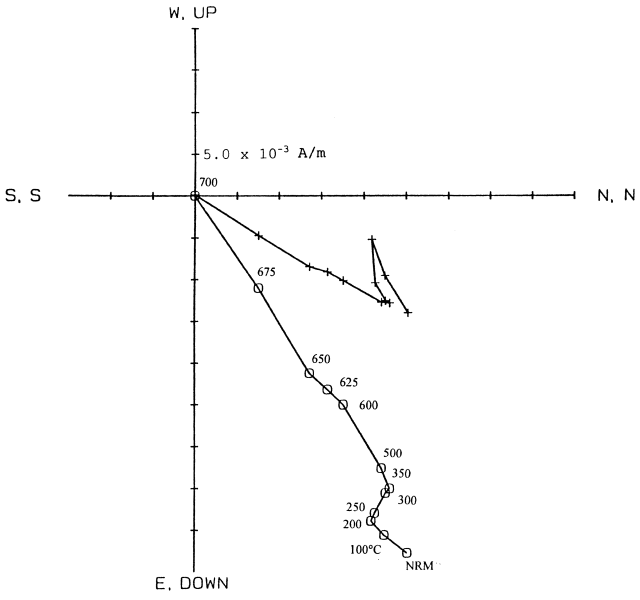


Figure 6. Representative vector plot of the thermal demagnetization of basalt sample 95K0043A of the Hakbong volcanic Formation.

4.1.3 *Late Aptian Haman Formation*

We sampled four red sandstone localities from outcrops at Sankyok-dong near the Kyongsang Provincial Building in Taegu. These sandstone rocks are in the lower part of the Haman Formation. Five red siltstone and sandstone localities on the campus of the Kyungpook National University were also sampled. These rocks are from the middle part of the Haman Formation. Stratigraphic and palaeontological studies as well as the new U–Pb age (113.6 ± 10 Ma) indicate a late Aptian age for these rocks, according to our field guides from the Kyungpook National University. We used progressive thermal demagnetization at a minimum of 10 levels to resolve characteristic components. Most samples from Sankyok-dong display rather straightforward demagnetization behaviour, characterized by a single linear trajectory over a broad temperature range after removal of a viscous component and sometimes a second ITC (Fig. 7a). The fact that the magnetization persists to temperature treatments of up to 675° indicates that haematite is probably the carrier of the ChRM. The ChRM directions are all normal, which may indicate that the magnetization was acquired during the Cretaceous Normal Superchron.

The magnetization of the red siltstone and sandstone from the Kyungpook National University campus is also relatively straightforward. A component with a direction aligned roughly along the present field direction at the sampling locality was typically removed by demagnetization treatments up to 500 °C. Further thermal demagnetization revealed in all samples a high unblocking temperature component of magnetization with a demagnetization trajectory that converges towards the origin of the vector plots (Fig. 7b). The direction of this final high temperature component is predominantly normal polarity, but four samples towards the bottom of the section revealed reversed polarity. The mean normal and reversed directions (Table 2) are not significantly different from antipodal at the 95 per cent confidence level (C class reversal test in McFadden & McElhinny 1990).

In contrast to other formations studied in the Taegu area which show persistent eastward-deflected declinations, the ChRM direction for the Haman Formation is directed more northerly with intermediate to steep downward inclination ($D = 14.6^\circ$, $I = 49.6^\circ$, $N = 9$ sites, $k = 64.0$, $\alpha 95 = 6.5^\circ$). After tectonic correction, the mean direction of the ChRM becomes $D = 16.2^\circ$, $I = 57.2^\circ$ with $k = 39.3$, and $\alpha 95 = 8.3^\circ$ (Table 2).

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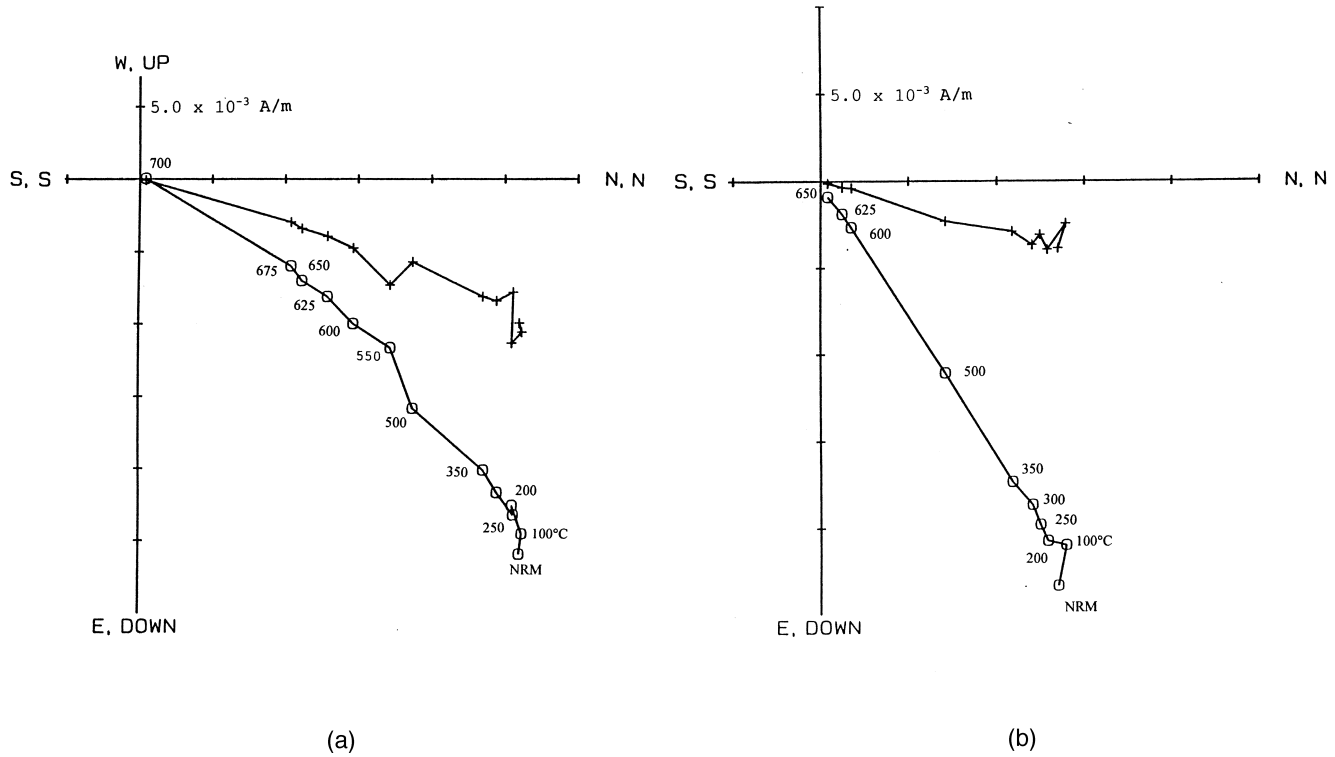


Figure 7. (a) Thermal demagnetization plots of red sandstone sample 95K0099B from the upper section of the Haman Formation. (b) Thermal demagnetization of red sandstone sample 95K0009A from the lower section of the Haman Formation. Directions are plotted in geographical coordinates. Crosses and circles as in Fig. 4.

4.2 Rock magnetic results

4.2.1 Hysteresis loop parameters

Hysteresis loop parameters are useful in characterizing the intrinsic magnetic behaviour of rocks and helpful in studying the origin of remanence. For the middle Aptian basalt samples, the slope of the isothermal remanence acquisition curve and ratios of saturation remanence and saturation magnetization (M_r/M_s) and remanence coercive force and coercive force (H_{cr}/H_c) are indicative of the presence of low-Ti titanomagnetite. On the other hand, the hysteresis loop for the late Aptian redbed sample displays a constricted loop (wasp-waisted), which is typical for the presence of low-coercivity magnetite and high-coercivity haematite (Opdyke & Channel 1996).

4.2.2 Curie temperature

We conducted the thermomagnetic analyses in an inert atmosphere and a magnetic field of 1 T on representative samples. The heating curves for the late Aptian redbed samples indicate that haematite could be responsible for much of the observed natural remanent magnetization.

4.2.3 Low-temperature properties

Low-temperature measurements were made on representative redbed samples to characterize further the magnetic minerals and enable their rock magnetic properties to be understood. As shown in Fig. 8, the low-temperature curves display a variety of features. These include a step-like loss of remanence in the vicinity of 120 K, which is caused by the Verwey transition of magnetite (Verwey *et al.* 1947). The general loss of remanence, excluding the Verwey step transition, reflects the presence of superparamagnetic grains which, due to thermal instabilities (Cullity 1972), are unable to retain remanence except at very low temperatures (Banerjee *et al.* 1993).

To sum up, the combined investigation suggests that mainly haematite and smaller amounts of magnetite control the magnetic properties of the late Aptian redbeds. The middle Aptian basalt contains titanomagnetite with variable particle size. The hysteresis ratios plotted on a Day *et al.* (1977)-type diagram suggest that the bulk magnetic grain size is in the pseudo-single domain size range (0.2–14 μm).

5 DISCUSSION

In summary, palaeomagnetic results from the Cretaceous rock samples for this study in the Kyongsang basin were magnetically straightforward with two distinct components of magnetization: (1) an intermediate unblocking temperature component (ITC), resembling the geocentric axial dipole field direction at the sampling localities, and (2) a higher unblocking temperature component (HTC), well defined in vector plots, whose remanance resides in haematite for the redbeds and in both magnetite and haematite for the basalts. The directions for the early to middle Aptian formations are generally directed to the northeast, whereas those of the late Aptian (the Haman Formation) are northerly, indicating some clockwise deflection of the directions between these Cretaceous strata. Thus, our data are in agreement with previous palaeomagnetic data from South Korea, and suggest that rocks from the southeast part

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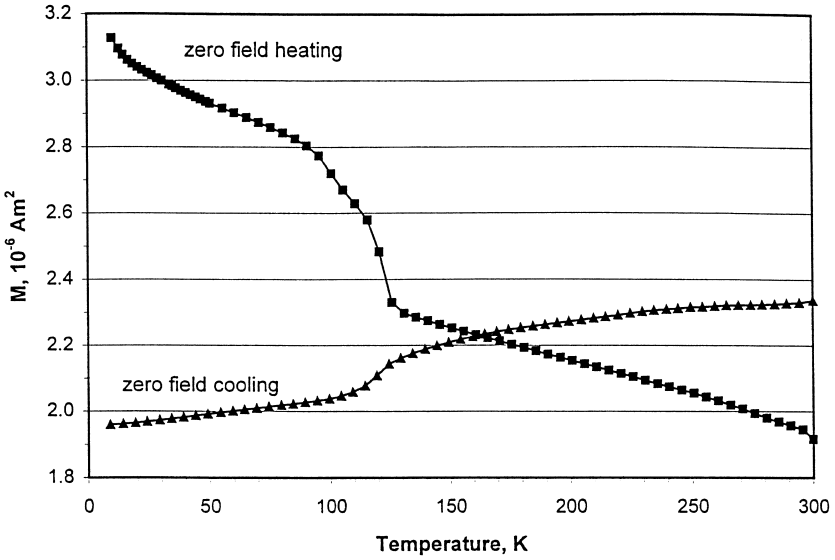


Figure 8. Low-temperature heating and cooling curves of saturation remanence normalized to 10 K. Triangles represent saturating the sample (95K011A) in a steady magnetic field of 2.5 T at room temperature and measuring the remanence at 5 K intervals while cooling down to 10 K in zero field; squares plot the saturation isothermal remanence at 10 K and then warming it to 300 K in zero field.

of the Okchon zone may have undergone clockwise rotation between the early and late Aptian. We will return to these results at the end of this section after a brief discussion about the age of magnetization.

5.1 Age of magnetization

Determining the age of magnetization is absolutely critical but not always straightforward. This has been the single most challenging problem in palaeomagnetic research in Asia. Fold tests have also been very difficult to obtain. The amount of tilt is typically small, so *in situ* palaeomagnetic vectors are not greatly changed by application of a tilt correction. Thus, many fold and tilt tests are statistically inconclusive. This problem was also manifested in some of our own data in this study: shallow dips (less than 20°) preclude a meaningful local fold test for most of the Cretaceous strata from the Taegu area.

The directions associated with the ChRM in the early Aptian red sandstones of the Silla Formation displayed high consistency and were isolated after the present-day field overprinting component had been successfully removed, leading us to interpret them as primary (Table 2). In addition, although presented only by a small percentage of samples, the ITC component with reversed polarity suggests that the ChRM was acquired before 0.78 Ma.

The characteristic directions of the early to middle Aptian basaltic rocks were isolated after removal of a soft secondary overprint. The characteristic magnetization in most basalts is carried by both magnetite and haematite, indicating that the haematite was probably formed by auto-oxidation processes during primary cooling of the volcanic rocks. There is no independent proof of this timing, but the ideal magnetization behaviour depicted in Fig. 6 shows no evidence of later disturbance of the ChRM. For these reasons, plus that of the straightforward demagnetization behaviour, we believe that the characteristic components can be regarded as recorders of the palaeofield sometime in the early to middle Aptian.

The results from the Haman Formation have both normal and reversed directions. The reversal test is positive, as mentioned above, indicating that the data probably give a reasonable time-averaged direction and that sufficient time has lapsed during the acquisition of the magnetization for the field to have reversed. Because the present-day field overprint component has been successfully removed and the pole position corresponding to the characteristic component falls close to that of the coeval strata seen elsewhere in the Kyongsang basin, the characteristic component isolated from the late Aptian rocks in this study is probably also primary. In addition, the same ChRM directions have been reported in previous studies (e.g. Otofuji *et al.* 1986) of the same formation in the same general area (see Table 2). Considering all the evidence, we interpret the characteristic component from the Haman Formation as a record of the palaeomagnetic field close in age to the deposition of these late Aptian sandstones.

Assuming that the above are correctly estimated, several critical questions still remain. Could the rotations be of larger extent other than the local rotations mentioned above within the Kyongsang basin? Are there other geological manifestations of the rotation that can be identified? What are the causes of the rotations? We compare our results with other available palaeomagnetic data to examine these questions below.

5.2 Comparison with other Cretaceous palaeomagnetic results from South Korea

Previous palaeomagnetic studies (with adequate magnetic cleaning techniques) on Cretaceous rocks include Ito & Tokieda (1980), Otofuji *et al.* (1986), Kim & Jeong (1986), Lee *et al.* (1987), Kim (1988), Doh & Piper (1994), and Lee *et al.* (1997). We have recalculated their data according to the age of the formations and re-analysed them in terms of a combined fold test (Table 2). Although bedding attitudes were not always available in the published work, an incremental fold test (Watson & Enkin 1993) was applied to the data set obtained from this study and those of Otofuji *et al.* (1986)

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(see Table 2). The concentration parameter maximizes at 100 per cent unfolding, indicating a positive fold test. Several recalculated data also pass the regional fold test at more than 95 per cent confidence (see $k2/k1$ ratios in Table 2), suggesting that the characteristic remanent magnetization (ChRM) is pre-tilting and probably primary. The mean declination calculated from the combined Early Cretaceous data set is 36.5° (Table 2). The declinations calculated for the early, middle, and late Aptian data sets are 45.9°, 38.1°, and 26.6°, respectively. Thus, this relative clockwise rotation is also suggested in the published Cretaceous poles, which were derived from widespread areas in southern Korea. It is interesting to note that the deflection pattern of the site mean directions is revealed in tilt-corrected coordinates and was not evident in the *in situ* orientation (Table 2). It is thus conceivable that the rotation pattern may be merely a coincidence of tilt correction. However, the mean ChRM directions were not remagnetized in recent times, as demonstrated by the positive fold and reversal tests, discussed above. The localities sampled are apparently structurally simple, and the tilt-corrected results are consistent for all studies. In addition, this rotation pattern apparently correlates with variations in the regional geological and topographic trends between these formations in Taegu (Fig. 3). The structural trends for the early and middle Aptian Formations are bent clockwise and mainly in a northeasterly direction (Fig. 3). These observations on declination–structural trend correlation suggest that the tilt-corrected directions are valid indications of original magnetization. A recent extensive study of Cretaceous rocks from South Korea by Kang & Kim (1998) further supports this notion. They collected nearly 2000 independently orientated core samples from seven widely separated areas in the Kyongsang basin. The ChRM obtained from both sedimentary and igneous rocks passes fold and reverse tests and the directions agree with ours very well.

As far as the timing of the rotations is concerned, we note the mean directions of the late Aptian Haman Formation in Kyongsang basin and the 73–82 Ma granite from Pulganshan are not significantly rotated (see Table 2). Therefore, it is likely that most or all of the relative rotations occurred during major regional deformation in the Early Cretaceous and may have ended before the Late Cretaceous. This time of completion for the rotations appears to be supported by the studies of Doh *et al.* (1998, personal communication) on Late Cretaceous strata from the northern boundary of the Okchon belt, where no detectable rotations in Late Cretaceous to Early Tertiary rocks were found. To analyse the regional tectonic implications further and to discuss mechanisms that could have produced the rotation during Cretaceous times, we next compare the Korean data with those of neighbouring blocks.

5.3 Comparison with Cretaceous poles from other blocks of Eastern Asia

We had already collected and examined all available Cretaceous poles from Siberia (SIB) and the NCB and SCB (Zhao *et al.* 1996; Gilder & Courtillot 1997). The mean Cretaceous poles for each block are listed in Table 3. The data-selection criteria that we used were similar to those generally used (e.g. Van der Voo 1990; Besse & Courtillot 1991; Beck 1998), with an emphasis on reasonably accurate age dating and on the evidence for a primary magnetization constrained by field tests (such as fold, reversal, and conglomerate tests).

In Table 3, the early (A1), middle (A2), and late Aptian (A3) palaeopoles for Korea (Table 2) are compared with Early Cretaceous palaeopoles for Siberia, the NCB and the SCB. Here the NCB (see Fig. 1) is defined as the landmass north of the Qinling suture and west of the Tanlu fault, and also includes the Mongolian blocks because they were accreted long before the Cretaceous. It is obvious from the data in Table 3 that Korea and other major parts of eastern Asia occupied the same relative positions in terms of palaeolatitudes in the Cretaceous. For example, the mean palaeolatitude predicted for Taegu (Table 3) is 38° within a 95 per cent confidence band of only $\pm 5°$. The palaeomagnetically inferred latitudinal displacements (F-values in Table 3) are general statistically insignificant and the positive (northward) and negative (southward) magnitudes are not significantly different from zero. The difference is only in terms of relative rotations as evidenced by the fact that the Korean pole positions are displaced eastwards with respect to the contemporaneous Eurasian poles (Table 3). The mean palaeomagnetic declinations for the Early and Late Cretaceous for Korea are deflected clockwise from the expected directions for the stable NCB, SIB, and SCB. Clockwise vertical-axis block rotations of $29.9° \pm 9.9°$ for the early Aptian, $24.9° \pm 10.6°$ for the middle Aptian, and little to none ($10.6° \pm 14.8°$) for the late Aptian are implied by the data (Table 3 and Fig. 9).

The above interpretation is compatible with several geological and geophysical observations. The active east Asian margin has experienced tectonic folding and faulting, widespread magmatic activity and remagnetization, and extensional basin formation (Li 1998). There were also changes in the directions of regional stress field: from northwesterly compression in the Early Cretaceous to easterly extension in the Late Cretaceous (Liu & Shan 1995). A recent report on seismic data from a survey in eastern China, in which a joint Chinese–German research team surveyed possible sites for a super deep borehole, suggests that the Tanlu fault is a near-vertical fault extending down to the Moho. In the entire crust, no seismic structures/reflectors migrate into regions east of the Tanlu fault (Schulze *et al.* 1998). Thus, it appears that, in this region of eastern Asia, crust perhaps breaks up into relatively small independent tectonic domains in which average clockwise rotation is imposed by the gross dextral shearing of Pacific–China plate interactions.

Although the palaeomagnetic investigations in east Asia are not sufficiently detailed to consider the problem of rotation mechanism, we note that the rotation appears to be closely related in space and time to the period of anomalously high convergence and subduction between the Kula oceanic plate and the east Asian continental plate in the Early Cretaceous (Engebretson *et al.* 1986; Rea & Duncan 1986). Thus, the Kula–Eurasia convergence could have been the motive force, and several published rotation mechanisms, such as the 'Domino-style' block rotations of Randall *et al.* (1996) and the 'Buttressed slivers' mechanism of Beck (1998), may have operated simultaneously to account for the rotation.

5.4 Tectonic affiliation of the Korean Peninsula

As mentioned earlier, this question is one of the highly charged controversies in Asian tectonics. The lack of knowledge about this question hampers our understanding concerning the models of collisions of eastern Asian blocks and their palaeogeographic

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Table 3. Cretaceous palaeomagnetic poles, palaeolatitudes, and tectonic parameters for East Asian blocks.

Block	VGP		Source	Plat	del-lat	Dec	Inc	del-d	Tectonic Parameters	
	Long/Lat	A_{95}							R del-R	F del-F
Early Cretaceous reference poles										
NCB	202.6/78.6	6.2	[1]	38.4	4.8	14.0	57.8	6.2		
SCB	219.6/77.3	7.0	[2]	34.5	5.5	21.3	54.0	6.6		
SIB	207.4/75.2	4.0	[3]	37.6	3.1	18.4	57.0	3.9		
(95–130 Ma)										
E. ASIA	210.0/77.1	3.9	[4]	37.0	3.0	16.0	56.4	3.8		
KOR: Early Aptian										
HTC	191.7/51.2	5.4	Table 2	43.7	4.2	50.3	62.4	5.8		
	Sampling area in this study with respect to NCB								36.3 ± 8.5	-5.3 ± 6.4
	Sampling area in this study with respect to SCB								29.0 ± 8.8	-9.2 ± 6.9
	Sampling area in this study with respect to SIB								31.9 ± 7.0	-6.1 ± 5.2
	Sampling area in this study with respect to E. ASIA								34.3 ± 6.9	-6.7 ± 5.2
Average										
	193.8/54.8	8.6	Table 2	42.8	6.7	45.9	60.7	9.2		
	South Korea with respect to E. ASIA								29.9 ± 9.9	-5.8 ± 7.3
KOR: Middle Aptian										
HTC	204.8/57.4	10.1	Table 2	37.0	7.9	40.9	56.5	9.9		
	Sampling area in this study with respect to NCB								26.9 ± 11.7	1.4 ± 9.2
	Sampling area in this study with respect to SCB								19.6 ± 11.9	-2.5 ± 9.6
	Sampling area in this study with respect to SIB								22.5 ± 10.6	0.6 ± 8.5
	Sampling area in this study with respect to E. ASIA								24.9 ± 10.6	0.0 ± 8.4
Average										
	205.4/59.6	11.4	Table 2	37.1	8.9	38.1	56.5	11.2		
	South Korea with respect to E. ASIA								22.1 ± 12.0	-0.1 ± 9.4
KOR: Late Aptian										
HTC	202.2/77.7	11.2	Table 2	38.6	8.7	15.1	58.0	11.2		
	Sampling area in this study with respect to NCB								1.1 ± 12.8	-0.2 ± 9.9
	Sampling area in this study with respect to SCB								-6.2 ± 13.0	-4.1 ± 10.3
	Sampling area in this study with respect to SIB								-3.3 ± 11.9	-1.0 ± 9.2
	Sampling area in this study with respect to E. ASIA								-0.9 ± 11.8	1.6 ± 9.2
Average										
	210.2/68.6	14.7	Table 2	36.3	11.5	26.6	55.8	14.3		
	South Korea with respect to E. ASIA								10.6 ± 14.8	-0.7 ± 11.9
KOR: Early Aptian–Late Aptian										
Average										
	199.5/61.2	6.6	Table 2	40.2	5.2	36.5	59.4	6.8		
	South Korea with respect to E. ASIA								20.5 ± 7.8	3.2 ± 6.0

Block: NCB = North China Block, SCB = South China Block, SIB = Siberia, KOR = Korea. Lat, Long: latitude and longitude of the north-seeking pole positions; A_{95}: radius of 95 per cent confidence circle of the pole. Source: [1] Gilder & Courtillot (1997); [2] Zhao *et al.* (1996); [3] Besse & Courtillot (1991); [4] Fisherian average of NCB, SCB, and SIB by this study. Dec, Inc: declination and inclination reduced at the sampling site (36°N, 129°E), respectively; del-lat, del-d, 95 per cent confidence limit in palaeolatitude and declination, respectively, del-lat = C × A95, C = 0.78, del-d = C $\sin^{-1}$ (sin [A_{95}]/cos [Plat]); Tectonic parameters—F and R: displacement northwards (+) or southwards (−) and azimuthal rotation clockwise (+) or counterclockwise (−) of sampling area with respect to reference blocks, inferred from the difference between the mean palaeomagnetic pole and the coeval reference pole. del-F and del-R: uncertainties (95 per cent confidence limits) of F and R, respectively, estimated by the method of Coe *et al.* (1985).

settings. The new Cretaceous data from this study have thrown some light on this controversy and allow us to re-examine this question with the available palaeomagnetic data from Korea and the North and South China blocks.

As shown in Table 4, the pre-Cretaceous palaeomagnetic poles for Korea are all different from each other, which makes it difficult to compare the coeval poles from the North and South China blocks. At first glance, Triassic and Jurassic palaeomagnetic poles for Korea published prior to 1994 (Shibuya *et al.* 1988; Kim & Van der Voo 1990; Kim *et al.* 1992) are indeed somewhat similar to coeval ones from the SCB, and this was perhaps the reason that led some workers to suggest that Korea and South China may have behaved as a single tectonic block since the Triassic (Kim *et al.* 1992; Gilder *et al.* 1995). A recent study of Middle and Late Carboniferous rock by Lee *et al.* (1996) added support to this hypothesis because the Carboniferous poles for Korea and the SCB are also in agreement (Table 4).

We argue, however, that all these poles that apparently correspond to the SCB poles were derived from rocks within the Okchon zone, which is known to be a site of severe deformation in the Mesozoic. In fact, in a re-study of the

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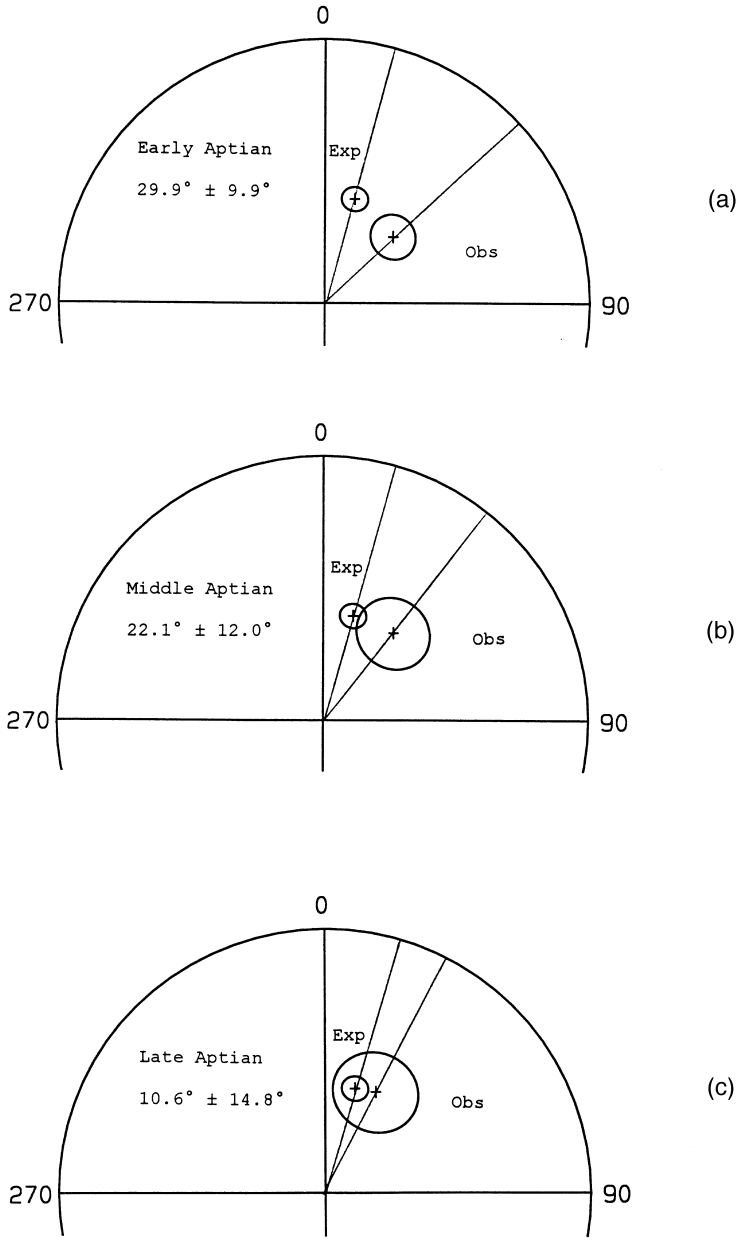


Figure 9. Equal-area projection of magnetic directions with 95 per cent confidence circles for South Korea listed in Table 3. (a) Early Aptian, (b) middle Aptian, (c) late Aptian. Obs: observed; Exp: expected directions reduced from the Eurasian poles.

same rocks Doh *et al.* (1997; p. 1228) concluded that the Carboniferous results by Lee *et al.* (1996) were Cretaceous remagnetization. Although detailed structural analysis and mapping are still insufficient to unravel the kinematic history of the Okchon foldbelt and assess its effects on these palaeomagnetic poles, we think that the proximity of these poles to the SCB poles may be a coincidence. Thus, we are inclined to leave out the poles derived from the Okchon zone (marked with stars in Table 4) at this time and merely retain those poles derived from areas bordering the Okchon zone (the underlined data in Table 4). If this assessment is correct, a striking feature between the Korean and NCB poles emerges: the Late Permian, Early Triassic, and Late Jurassic poles for Korea are systematically displaced some 30° eastwards with respect to the coeval poles of the NCB. A 30° clockwise rotation about a vertical or near-vertical axis brings the Korean poles into general coincidence with the coeval poles for the NCB. This suggests that the Late Palaeozoic and Mesozoic poles for Korea may all have undergone the Cretaceous-aged clockwise rotation deduced from this study, or, in other words, that the whole region rotated clockwise by about 30° with respect to the NCB during the Cretaceous. As shown in Table 4 and Fig. 10, a remarkable coincidence of these poles for Korea and the NCB is achieved by adjusting the 30° clockwise rotations to the Korean poles. Although the Ordovician and Carboniferous poles for the NCB and Korea are quite different from each other, their confidence circles still overlap (Fig. 10b). Korea and the NCB therefore may have been part of the same continental landmass since at least Late Permian times, probably even from the Early Palaeozoic.

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Table 4. Published palaeomagnetic poles for Korea, and the North and South China blocks.

Age	Korea Long/Lat	A95	Ref	Corrected Long/Lat	NCB Long/Lat	A95	Ref	SCB Long/Lat	A95	Ref
O1 (509 ± 2 Ma)	169.2°E/−59.9°N	17.5°	[1]	202.9°E/−39.3°N	176.1°E/−43.6°N	8.5°	[2]	22.0°E/−17.0°N	3.0°	[3]
C3	335.0°E/44.6°N	6.9°	[4]	6.5°E/28.3°N	354.8°E/44.1°N	18.6°	[5]	227.1°E/19.1°N	16.1°	[6]
	*226.8°E/12.2°N	15.5°	[1]							
	*147.6°E/−5.7°N	6.9°	[7]							
	*220.2°E/−15.3°N	13.0°	[8]							
P2	311.9°E/58.7°N	4.1°	[4]	359.3°E/47.4°N	357.5°E/49.1°N	6.9°	[9]	242.5°E/52.9°N	7.2°	[9]
	*231.0°E/5.7°N	16.7°	[8]							
	*203.1°E/9.4°N	19.7°	[1]							
	*127.5°E/40.6°N	8.5°	[7]							
T1	306.1°E/63.2°N	12.6°	[4]	0.3°E/52.6°N	357.4°E/57.9°N	4.8°	[9]	218.4°E/45.6°N	6.5°	[9]
	*209.1°E/22.6°N	10.4°	[8]							
	*215.9°E/33.5°N	16.3°	[10]							
	*179.0°E/24.6°N	4.7°	[7]							
J3	199.3°E/59.5°N	12.8°	[10]	185.5°E/82.0°N	222.8°E/74.4°N	5.9°	[11]	211.4°E/74.9°N	10.3°	[11]

Age: O, Ordovician; C, Carboniferous, P, Permian, T, Triassic, J, Jurassic; 1 = Early, 2 = Middle 3 = Late.
Ref: [1] Lee *et al.* (1997), [2] Frost (1994), [3] Li (1988), [4] Doh & Piper (1994), [5] Wu (1988), [6] Lin *et al.* (1985), [7] Shibuya *et al.* (1988), [8] Kim *et al.* (1992), [9] Zhao *et al.* (1996), [10] Kim & Van der Voo (1990), [11] Gilder & Courtillot (1997).
* Palaeomagnetic poles derived from the Okchon zone. Underlined data are used in Fig. 10. Corrected: obtained by applying the 30° correction to the corresponding declination reduced at the Taegu area and then recalculating the pole. Other symbols are as in Tables 2 and 3.

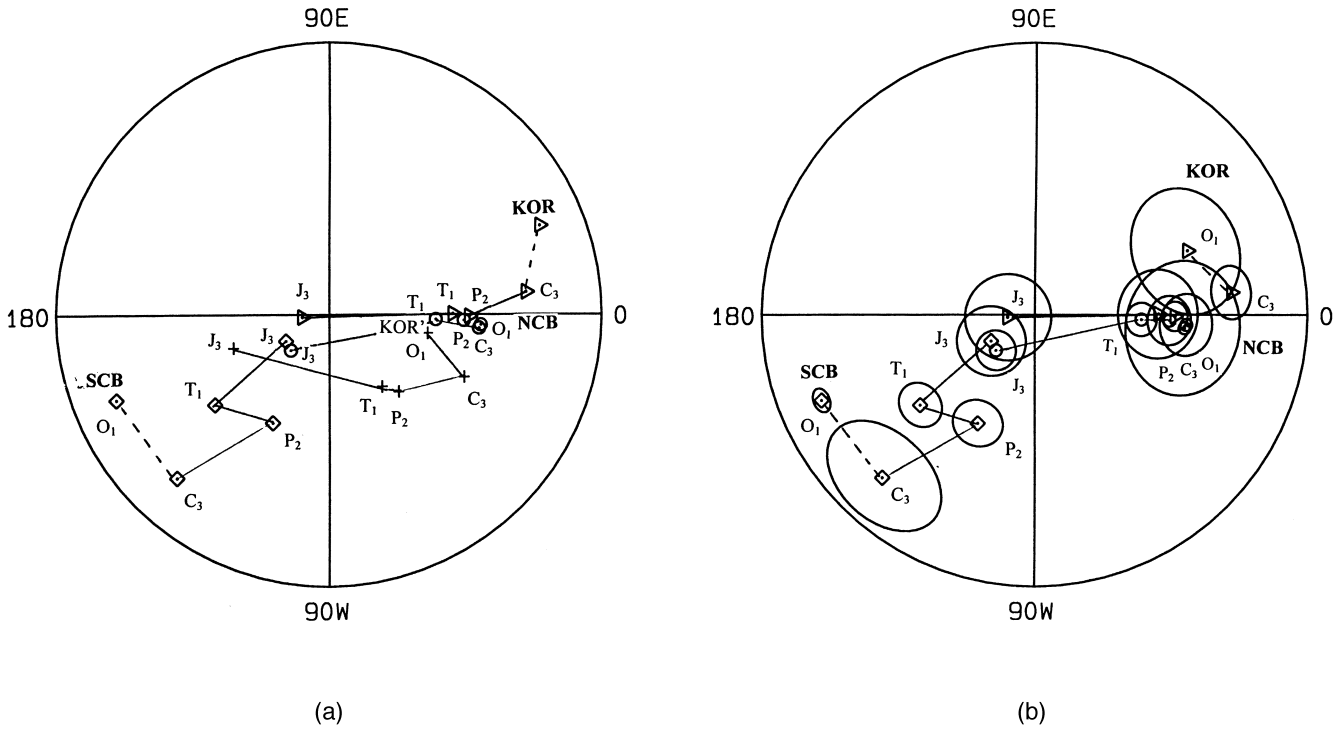


Figure 10. Equal-area projection of palaeomagnetic poles (a) without and (b) with 95 per cent confidence circles for the North and South China blocks and Korea (Table 4), displaying the position of an unrotated KOR polar path with respect to the NCB and SCB polar paths. The triangle KOR curves are plotted adjusted for the 30° clockwise rotation. The Early Ordovician poles were inverted and plotted for the purpose of clarity.

It is apparent in Fig. 10 that the rotated (or corrected) poles show a large APW motion of Korea between the Early Triassic and Late Jurassic, replicating those found in the palaeomagnetic results from the NCB (Zhao *et al.* 1990, 1996; Gilder & Courtillot 1997). Late Jurassic palaeomagnetic poles for the NCB, SCB, and Korea are statistically indistinguishable, reinforcing the hypothesis that the accretion of the North and South China blocks was finished at this time (Zhao & Coe 1987; Gilder & Courtillot 1997). Thus, the NCB–Korea connection is not only consistent with the majority of geological

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observations, suggesting affinities between the two blocks, but is also consistent with the collisional tectonic history of the eastern Asian margin derived from palaeomagnetic data.

5.5 Early Cretaceous palaeogeographic reconstruction of East Asia

To synthesize our current palaeomagnetic knowledge for eastern Asia, we have attempted a palaeogeographic reconstruction of the Early Cretaceous in the East Asian margin. For the sake of simplicity, we only emphasize Siberia, and the NCB and SCB in the reconstruction (Fig. 11). These blocks are positioned according to the mean palaeopole in Table 3, with their approximate Cretaceous palaeolatitudes and uncertainties indicated. Note that these blocks can be moved along lines of equal latitude because the palaeolongitudes are unconstrained by palaeomagnetic data. We have taken into account those geological observations that we think are most relevant. As shown in Fig. 11, the eastern Asian continent reaches a simple bulky shape that is retained until upsets by the Pacific–Eurasia plate convergence during the Late Mesozoic and by the collision of India in the Early Tertiary. Our reconstruction is very similar to those in previous works (such as Chen *et al.* 1993; Halim *et al.* 1998). The reconstruction in Fig. 11 differs from previous ones mainly in the addition of new palaeomagnetic results from Korea. We have indicated the areas in southern Korea that are to be rotated clockwise about a vertical axis from the expected Cretaceous directions. Much work still needs to be done, both palaeomagnetic and geological, to better constrain the extent of rotation within Korea and around the region east of the Tanlu fault (Gilder *et al.* 1999).

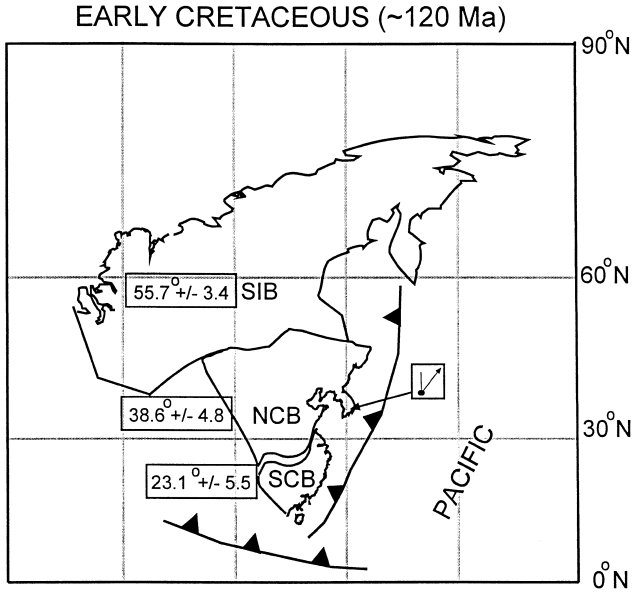


Figure 11. Early Cretaceous (~120 Ma) palaeogeographic map showing possible positions of the three major East Asian blocks (abbreviations as in Fig. 1). Palaeolatitudes and 95 per cent confidence limits (estimated by the method of Coe *et al.* 1985) for the central positions of SIB (60°N, 110°E), NCB (40°N, 110°E) and SCB (28°N, 110°E) are shown according to the mean palaeomagnetic poles selected in Table 3. The box indicates regions in southern Korea that are to be rotated 30° clockwise about a vertical or near vertical-axis.

6 CONCLUSIONS

Relative clockwise rotations of up to 36° between South Korea and the stable NCB are demonstrated by palaeomagnetic data of Cretaceous age in Korea. The new Cretaceous palaeomagnetic data from this study are consistent with the hypotheses that Korea and other major parts of eastern Asia occupied the same relative positions in terms of palaeolatitudes in the Cretaceous, and that regions south of the Okchon zone experienced vertical or near-vertical axis clockwise rotation between 121 and 114 Ma (Aptian) in the middle part of Early Cretaceous. The clockwise rotation was probably caused by the Kula–Eurasia plate convergence during the Late Mesozoic. An analysis and comparison of previously reported palaeomagnetic data corroborates this hypothesis and suggests that much of Korea may have been connected to the North China Block prior to the Cretaceous.

ACKNOWLEDGMENTS

Discussions with our many colleagues have been most stimulating and helpful. We particularly wish to acknowledge the expertise, helpful advice, and continuing discussions generously given by Drs Zhengxiang Li, Kainian Huang, Neil Opdyke, Randy Enkin, Zhengyu Yang, Seong-Jae Doh and Yo-ichiro Otofuji. During various stages of this research, palaeomagnetic software developed by Randy Enkin was used. We thank Franz Heider of the journal editorial board, Randy Enkin and an anonymous reviewer for constructive suggestions on the original manuscript. We are pleased to acknowledge the support of the US National Science Foundation grant EAR-9419269, a grant from the California Space Institute, and the Chinese NSF grant 49334050. This manuscript is contribution 350 of the Institute of Tectonics and Palaeomagnetism Laboratory at the University of California, Santa Cruz.

REFERENCES

Banerjee, S.K., Hunt, C.P. & Liu, X.M., 1993. Separation of local signals from the regional paleomonsoon record of the Chinese Loess Plateau: a rock-magnetic approach, *Geophys. Res. Lett.,* **20,** 843–846.

Beck, M.E., 1976. Discordant paleomagnetic pole position as evidence of regional shear in the western cordillera of North America, *Am. J. Sci.,* **276,** 694–712.

Beck, M.E., 1998. On the mechanism of crustal block rotations in the central Andes, *Tectonophysics,* **299,** 75–92.

Besse, J. & Courtillot, V., 1991. Revised and synthetic apparent polar wander paths of the African, Eurasian, North American, and Indian plate and true polar wander since 200 Ma, *J. geophys. Res.,* **96,** 4029–4050.

Chang, K.H., 1988. Cretaceous stratigraphy and paleocurrent analysis of Kyongsang Basin, Korea, *J. geol. Soc. Korea,* **24,** 194–205.

Chang, K.H., 1995. Aspects of geologic history of Korea, *J. geol. Soc. Korea,* **31,** 72–90.

Chang, K.H. & Park, S.O., 1995. Cretaceous stratigraphy and geologic history of Taegu-Kyongju area, Korea, *Proc. 15th Int. Symp. Kyungpook National University,* pp. 419–434, eds Chang, K.H. & Park, S.O.

Chang, K.H., Lee, Y.J., Suzuki, K. & Park, S.O., 1998. Zircon morphology, CHIME age and geological significance of Kusandong Tuff (Cretaceous), *J. geol. Soc. Korea,* **34,** 333–342.

Chen, Y., Courtillot, V., Cogne, J.P., Besse, J., Yang, Z. & Enkin, R., 1993. The configuration of Asia prior to the collision of India: Cretaceous paleomagnetic constraints, *J. geophys. Res.,* **98,** 21 927–21 941.

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Coe, R.S., Globerman, B.R., Plumley, P.W. & Thrupp, G.A., 1985. Paleomagentic results from Alaska and their tectonic implications, in *Tectonostratigraphic Terranes of the Circum-Pacific Region,* pp. 85–108, ed. Howell, D.G., Am. Assoc. petrol. Geol., Circum-Pacific Council on Energy and Resources Earth Sciences Series No. 1.

Cullity, B.D., 1972. *Introduction to Magnetic Materials,* Addison-Wesley, Reading, MA.

Day, R., Fuller, M. & Schmidt, V.A., 1977. Hysteresis properties of titano-magnetites: grain-size and compositional dependence, *Phys. Earth planet. Inter.,* **13,** 260–267.

Doh, S.J. & Piper, J.D.A., 1994. Palaeomagnetism of the (upper Palaeozoic–lower Mesozoic) Pyongan Supergroup, Korea; a Phanerozoic link with the North China Block, *Geophys. J. Int.,* **117,** 850–863.

Doh, S.J., Suk, D.W. & Park, Y.H., 1997. Remagnetization of the Pyeongan Supergroup in the Yeongwol Area, Korea, *J. Geomag. Geoelectr.,* **49,** 1217–1232.

Engebretson, D.C., Cox, A. & Gordon, R.G., 1986. Relative motions between oceanic and continental plates in the Pacific basin, *Geol. Soc. Am. Spec. Paper* 206.

Enkin, R.J., Yang, Z., Chen, Y. & Courtillot, V., 1992. Palaeomagnetic constraints on the geodynamic history of the major blocks of China from the Permian to the present, *J. geophys. Res.,* **97,** 13 953–13 989.

Fisher, R.A., 1953. Dispersion on a sphere, *Proc. R. Soc. Lond. A,* **217,** 295–305.

Frost, G.M., 1994. Paleomagnetic studies on the tectonic evolution of north China and Qaidam, *PhD thesis,* University of California, Santa Cruz.

Gilder, S. & Courtillot, V., 1997. Timing of the North–south China collision from new Middle to Late Mesozoic paleomagnetic data from the North China Block, *J. geophys. Res.,* **102,** 17 713–17 727.

Gilder, S.A., Zhao, X.X., Coe, R.S., Wu, H.R. & Kuang, G.D., 1993. Discordance of Jurassic paleomagnetic data from south China and their tectonic implications, *Earth planet. Sci. Lett.,* **119,** 259–269.

Gilder, S.A., Coe, R.S., Wu, H.R., Kuang, G.D., Zhao, X.X. & Wu, Q., 1995. Triassic paleomagnetic data from southern China and their bearing on the tectonic evolution of the western circum-Pacific region, *Earth planet. Sci. Lett.,* **131,** 269–287.

Gilder, S., Leloup, P.H., Courtillot, V., Chen, Y., Coe, R., Zhao, X., Xiao, W., Halim, N., Cogn, J.P. & Zhu, R.X., 1999. Tectonic evolution of the Tancheng-Lujiang (Tan-Lu) Fault via Middle Triassic to Early Cenozoic paleomagnetic data, *J. geophys. Res.,* **104,** 15 365–15 390.

Halim, N., Kravchinsky, V., Gilder, S., Cogne, J.P., Alexyutin, M., Sorokin, A., Courtillot, V. & Chen, Y., 1998. A paleomagnetic study from the Mongol-Okhotsk region: rotated early Cretaceous volcanics and remagnetized Mesozoic sediments, *Earth planet. Sci. Lett.,* **159,** 133–145.

Huang, K.N. & Opdyke, N.D., 1991. Paleomagnetism of Jurassic rocks from southwestern Sichuan and the timing of the closure of the Qinling suture, *Tectonophysics,* **200,** 299–316.

Huang, K.N. & Opdyke, N.D., 1996. Severe remagnetization revealed from Triassic platform carbonates near Guiyang, Southwest China, *Earth Planet. Sci. Lett.,* **143,** 49–61.

Ito, H. & Tokieda, K., 1980. An interpretation of paleomagnetic results from Cretaceous granites in South Korea, *J. Geomag. Geoelectr.,* **32,** 275–284.

Kang, H.C. & Kim, I.S., 1998. Palaeomagnetism of the Cretaceous Kyongsang Supergroup and igneous rocks from the Kyongsang Basin, South Korea, *AGU 1998 Western Pacific Geophysics Meeting Abstract, EOS Trans. Am. geophys. Un. (Suppl.),* **79**(24), W110.

Kent, D.V., Zeng, X.S., Zhang, W.Y. & Opdyke, N.D., 1987. Widespread Late Mesozoic to recent remagnetization of Paleozoic and Lower Triassic sedimentary rocks from south China, *Tectonophysics,* **139,** 133–143.

Kim, I.S., 1988. Paleomagnetism of Cretaceous Tadaepo Formation, Pusan area: its stratigraphic and tectonic significances, *J. geol. Soc. Korea,* **24,** 211–233.

Kim, K.H. & Jeong, B.I., 1986. A study of the paleomagnetism of Southern Korea since Permian, *J. Kor. Inst. mining Geol.,* **19,** 67–83.

Kim, K.H. & Van der Voo, R., 1990. Jurassic and Triassic paleomagnetism of South Korea, *Tectonics,* **9,** 699–718.

Kim, K.H., Lee, J.D. & Choi, M.O., 1992. Carboniferous-Triassic paleomagnetism of South Korea, *J. Geomag. Geoelectr.,* **44,** 959–978.

Kirschvink, J.L., 1980. The least squares line and plane and the analysis of paleomagnetic data, *Geophys. J. R. astr. Soc.,* **62,** 699–718.

Kong, F.C., Lawver, L.A. & Lee, T.Y., 1997. Evolution of the East China Sea Basin and adjacent plate interaction (Abstract), *EOS. Trans. Am. geophys. Un.,* **78,** F177.

Lee, D.S., ed., 1987. *Geology of Korea,* Kyohak-Sa Publishing Co., Seoul.

Lee, G., Besse, J., Courtillot, V. & Montigny, R., 1987. Eastern Asia in the Cretaceous: new paleomagnetic data from South Korea and a new look at Chinese and Japanese data, *J. geophys. Res.,* **92,** 3580–3596.

Lee, Y.S., Nishimura, S. & Min, K.D., 1996. High unblocking temperature haematite magnetizations of Late Palaeozoic red beds from the Okchon zone, southern part of the Korean Peninsula, *Geophys. J. Int.,* **125,** 266–284.

Li, Z.X., 1988. Paleozoic paleomagnetism of Australia: tectonic significance and comparison with South China, *PhD thesis,* Macquarie University, Sydney

Li, Z.X., 1998. Tectonic history of the major east Asian lithospheric blocks since the mid-Proterozoic—A synthesis, in *Mantle Dynamics and Plate Interactions in East Asia,* pp. 221–243, eds Martin, F.J., Chung, S.-L., Lo, C.-J. & Lee, T.-Y., AGU Geodyn. Ser. 27, Am. geophys. Un., Washington, DC.

Lin, J.L., Fuller, M. & Zhang, W.Y., 1985. Preliminary Phanerozoic polar wander paths for the North and South China blocks, *Nature,* **313,** 444–449.

Liu, X.W. & Shan, J.R., 1995. Advances in studies of Tan-Lu fault zone and strike-slip fault system in northeast Asia, *Adv. Earth Sci.,* **10,** 176–182 (in Chinese with English Abstract),.

Luyendyk, B.P., Kammerling, M.J., Terres, R.R. & Hornafius, J.S., 1985. Simple shear of southern California during Neogene time suggested by paleomagnetic declinations, *J geophys Res.,* **90,** 12 454–12 466.

Maruyama, S., Li, J.G. & Seno, T., 1989. Mesozoic and Cenozoic evolution of Asia, in *The Evolution of the Pacific Ocean Margins,* pp. 75–99, ed. Ben-Avraham, Z., Oxford Monographs on Geology and Geophysics no. 8, Oxford.

McFadden, P.L. & McElhinny, M.W., 1990. Classification of the reversal test in palaeomagnetism, *Geophys. J. Int.,* **130,** 725–729.

Min, K.D., Lee, Y.S. & Hwang, S.Y., 1993. Paleomagnetic study of the Lower Ordovician Formations in the North Eastern Okchon Zone, *J. Kor. Inst. min. Geol.,* **26,** 395–401.

Opdyke, N.D. & Channel, J.E.T., 1996. *Magnetic Stratigraphy,* Academic Press, San Diego.

Otofuji, Y., Jin, Y.O., Hirajima, T., Kyung, D.M. & Sasajima, S., 1983. Paleomagnetism and age determination of Cretaceous rocks from Gyeongsang basin, Korean Peninsula, in *The Tectonic and Geologic Evolution of Southeast Asian Seas and Island, Part 2,* Geophys. Monogr. Ser., vol. 27, pp. 388–396, ed. Hayes, D., Am. geophys. Un., Washington, DC.

Otofuji, Y., Kim, K.H., Inokuchi, H., Morinaga, H., Murata, F., Katao, H. & Yaskawa, K., 1986. A paleomagnetic reconnaisance of Permian to Cretaceous sedimentary rocks in southern part of Korean Peninsula, *J. Geomag. Geoelectr.,* **38,** 387–402.

Otofuji, Y., Katsuragi, K., Inokuchi, H., Yaskawa, K., Kim, K.H., Lee, D.S. & Lee, H.Y., 1989. Remagnetization of Cambrian to Triassic sedimentary rocks of the Paegunsan Syncline of the Okch'on Zone, South Korea, *J. Geomag. Geoelectr.,* **41,** 119–135.

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Randall, D., Taylor, G. & Grocott, J., 1996. Major crustal rotations in the Andean margin: paleomagnetic results from the coastal cordillera of northern Chile, *J. geophys. Res.,* **101,** 15 783–15 798.

Rea, D.K. & Duncan, R.A., 1986. North Pacific plate convergence: a quantitative record of the past 140 m.y., *Geology,* **14,** 373–376.

Reedman, A.J. & Um, S.H., 1975. *Geology of Korea,* Korean Institute of Energy and Resources, Seoul.

Schulze, A., Jiang, M., Ryberg, T. & Gao, R., 1998. Survey yields data on unique metamorphic rock complex in China, *EOS, Trans. Am. geophys Un.,* **79,** 429–433.

Shibuya, H., Min, K.D., Lee, Y.S., Sasajima, S. & Nishimura, S., 1988. Paleomagnetism of Cambrian to Jurassic sedimentary rocks from the Ogcheon Zone, southern part of Korean Peninsula, *J. Geomag. Geoelectr.,* **40,** 1469–1480.

Van der Voo, R., 1990. Phanerozoic paleomagnetic poles from Europe and North America and comparisons with continental reconstructions, *Rev. Geophys.,* **28,** 167–206.

Verwey, E.J., Haayman, P.W. & Romeijn, F.C., 1947. Physical properties and cation arrangements of oxides with spinel structure, *J. Chem. Phys.,* **15,** 181–187.

Watson, G.S. & Enkin, R.J., 1993. The fold test in paleomagnetism as a parameter estimation problem, *Geophys. Res. Lett.,* **20,** 2135–2137.

Wu, H.N., 1988. Apparent polar wander paths and paleolatitude distributions for the North and South China blocks, the geotectonic evolution of Qinling belt, *PhD thesis,* Institute of Geology, Academia Sinica, Beijing (in Chinese with English abstract).

Xu, J.W., 1993. Basic characteristics and tectonic evolution of the Tancheng-Lujiang fault zone, in *The Tancheng-Lujiang Wrench Fault System,* pp. 17–50, ed. Xu, J.W., John Wiley and Sons Ltd, New York.

Xu, J.W., Zhu, G., Tong, W., Cui, K. & Liu, Q., 1987. Formation and evolution of the Tancheng-Lujiang wrench fault system: a major shear system of the Northwest Pacific Ocean, *Tectonophysics,* **134,** 273–310.

Yanai, S.C., Jwa, Y.J., Otoh, S., Yamakita, S. & Park, B.S., 1993. Honam Intra-Arc Transcurrent Fault and Jurassic Geodynamics in East Asia, in *The Tancheng-Lujiang Wrench Fault System,* pp. 213–224, ed. Xu, J.W., John Wiley and Sons Ltd, New York.

Yang, S.Y., 1984. Palaeontological study on the Molluscan Fauna from the Myogok Formation, Korea, *J. geol. Soc. Korea,* **20,** 15–27.

Yin, A. & Nie, S.Y., 1996. A Phanerozoic palinspastic reconstruction of China and its neighboring regions, in *Tectonic Evolution of Asia,* pp. 442–485, eds Yin, A. & Harrison, M., Cambridge University Press, Cambridge.

Zhao, X., 1987. A palaeomagnetic study of Phanerozoic rock units from eastern China, *PhD thesis,* University of California, Santa Cruz.

Zhao, X. & Coe, R.S., 1987. Palaeomagnetic constraints on the collision and rotation of North and South China, *Nature,* **327,** 141–144.

Zhao, X., Coe, R.S., Zhou, Y.X., Wu, H.R. & Wang, J., 1990. New palaeomagnetic results from northern China: collision and suturing with Siberia and Kazakhstan, *Tectonophysics,* **181,** 43–81.

Zhao, X., Coe, R., Gilder, S.A. & Frost, G.M., 1996. Palaeomagnetic constraints on the palaeogeography of China: implications for Gondwanaland, *Aust. J. Earth Sci.,* **43,** 643–672.

Zheng, Z., Kono, M., Tsunakawa, H., Kimura, G., Wei, Q.Y., Zhu, X.Y. & Hao, T., 1991. The apparent polar wander path for North China Block since the Jurassic, *Geophys. J. Int.,* **104,** 29–40.

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